GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Modulation Of Chemical Synaptic Transmis	−0.20569709	105	3.636e−13	1.959e−09	CPLX3:3 DRD2:28 DGKI:44 TRIO:52 GRM6:85 NPTX1:88
NADH Dehydrogenase Complex Assembly (GO:	0.29380758	48	1.969e−12	3.536e−09	NDUFB8:76 ECSIT:170 NDUFA9:278 TMEM126B:302 NDUFB6:335 TMEM126A:366
Mitochondrial Respiratory Chain Complex	0.29380758	48	1.969e−12	3.536e−09	NDUFB8:76 ECSIT:170 NDUFA9:278 TMEM126B:302 NDUFB6:335 TMEM126A:366
Chemical Synaptic Transmission (GO:00072	−0.12974537	237	7.453e−12	1.004e−08	GABRR1:14 DRD2:28 HRH3:53 RPS6KA2:63 NPTX1:88 SLC18A2:95
Mitochondrial Respiratory Chain Complex	0.21577959	77	6.264e−11	6.749e−08	NDUFB8:76 ECSIT:170 NDUFA9:278 TMEM126B:302 NDUFB6:335 TMEM126A:366
Mitochondrial Gene Expression (GO:014005	0.18623238	99	1.642e−10	1.474e−07	PTCD3:56 TFB2M:98 POLRMT:128 TFAM:342 MRPL37:412 MRPS30:469
Visual Perception (GO:0007601)	−0.18494442	90	1.410e−09	1.085e−06	TRPM1:5 CRYBB2:6 CRYAA:12 CRYBB1:40 VAX2:45 CRYBB3:66
Calcium Ion Transport (GO:0006816)	−0.17007061	102	3.144e−09	2.117e−06	TRPM1:5 TRPM5:35 ATP2A2:67 CACNA2D4:112 ITPR1:150 TRPM7:229
Mitochondrial Electron Transport, NADH T	0.30232565	31	5.777e−09	3.458e−06	NDUFB8:76 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471 NDUFB5:473
Mitochondrial ATP Synthesis Coupled Elec	0.23016349	52	9.666e−09	3.759e−06	NDUFB8:76 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471 NDUFB5:473
Monoatomic Cation Transmembrane Transpor	−0.10242826	269	8.810e−09	3.759e−06	TRPM1:5 TRPM5:35 KCNA10:51 ATP2A2:67 KCNC2:78 KCNQ3:100
Oxidative Phosphorylation (GO:0006119)	0.22604293	54	9.433e−09	3.759e−06	NDUFB8:76 ATP5PO:105 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471
Proton Motive Force−Driven ATP Synthesis	0.23232710	51	9.768e−09	3.759e−06	NDUFB8:76 ATP5PO:105 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471
Sensory Perception Of Light Stimulus (GO	−0.17530134	91	7.920e−09	3.759e−06	TRPM1:5 CRYBB2:6 CRYAA:12 CRYBB1:40 VAX2:45 CRYBB3:66
Anterograde Trans−Synaptic Signaling (GO	−0.12720026	169	1.296e−08	4.493e−06	GABRR1:14 HRH3:53 RPS6KA2:63 NPTX1:88 SLC18A2:95 KCNQ3:100
Mitochondrial Translation (GO:0032543)	0.16896726	95	1.334e−08	4.493e−06	PTCD3:56 MRPL37:412 MRPS30:469 GATB:476 MRPS23:559 MRPL9:732
Proton Motive Force−Driven Mitochondrial	0.23709897	47	1.918e−08	6.077e−06	NDUFB8:76 ATP5PO:105 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471
Muscle Contraction (GO:0006936)	−0.17539063	84	2.885e−08	8.634e−06	MYH13:198 NOS1:341 MYOM2:344 LMOD2:407 DES:443 KCNJ12:469
Aerobic Electron Transport Chain (GO:001	0.22086990	52	3.695e−08	1.039e−05	NDUFB8:76 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471 NDUFB5:473
Potassium Ion Transmembrane Transport (G	−0.13996686	130	3.856e−08	1.039e−05	KCNA10:51 KCNC2:78 KCNQ3:100 KCNQ2:125 KCNS3:129 KCNJ5:168
Regulation Of Synaptic Transmission, Glu	−0.21493221	47	3.508e−07	8.998e−05	DRD2:28 DGKI:44 GRM6:85 MAPK8IP2:132 TSHZ3:179 GRM8:204
tRNA Methylation (GO:0030488)	0.23999558	36	6.347e−07	1.554e−04	THADA:13 TRMT1:91 TRMT10A:210 FTSJ1:451 TRMT10C:645 TRMT9B:674
Alpha−Amino Acid Catabolic Process (GO:1	0.26535464	29	7.662e−07	1.794e−04	HIBCH:324 HMGCL:417 ALDH4A1:456 TAT:1107 KMO:1179 GOT2:1238
Aerobic Respiration (GO:0009060)	0.19580357	53	8.367e−07	1.878e−04	NDUFB8:76 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471 NDUFB5:473
Positive Regulation Of Synaptic Transmis	−0.17626788	65	9.153e−07	1.972e−04	SHANK2:11 TSHZ3:179 CACNG5:205 PTK2B:273 SYT1:398 STX4:487
Potassium Ion Transport (GO:0006813)	−0.13042104	116	1.288e−06	2.668e−04	KCNA10:51 KCNC2:78 KCNQ3:100 KCNQ2:125 KCNS3:129 KCNJ5:168
Inorganic Cation Transmembrane Transport	−0.08518007	272	1.513e−06	3.019e−04	TRPM1:5 TRPM5:35 KCNA10:51 ATP2A2:67 KCNC2:78 KCNQ3:100
Calcium Ion Transmembrane Transport (GO:	−0.15677935	78	1.750e−06	3.367e−04	TRPM1:5 TRPM5:35 ATP2A2:67 CACNA2D4:112 TRPM7:229 LETM1:411
Striated Muscle Contraction (GO:0006941)	−0.19380745	50	2.169e−06	4.030e−04	TNNI2:222 CSRP3:322 NOS1:341 RYR2:505 TNNT1:922 TCAP:939
Cardiac Conduction (GO:0061337)	−0.21243957	41	2.555e−06	4.588e−04	BIN1:124 KCNJ5:168 DSC2:190 CAMK2D:287 DSG2:430 MEF2A:687
Metal Ion Transport (GO:0030001)	−0.11194512	148	2.764e−06	4.803e−04	KCNA10:51 KCNS3:129 ITPR1:150 KCNJ5:168 TRPM7:229 SCN2A:240
Fatty Acid Oxidation (GO:0019395)	0.19710159	47	2.993e−06	4.885e−04	PHYH:16 ACADM:17 CPT2:116 HAO2:207 EHHADH:216 ECI2:304
Mitochondrial RNA Metabolic Process (GO:	0.30967997	19	2.982e−06	4.885e−04	TFB2M:98 POLRMT:128 TFAM:342 TEFM:507 FASTKD5:648 FASTKD1:713
Fatty Acid Beta−Oxidation (GO:0006635)	0.19652999	47	3.196e−06	5.065e−04	ACADM:17 ACOX2:38 CPT2:116 EHHADH:216 ECI2:304 GCDH:378
tRNA Modification (GO:0006400)	0.16444828	66	3.936e−06	6.057e−04	THADA:13 TRMT1:91 TRMT10A:210 FTSJ1:451 DTWD1:532 TRMT9B:674
Calcium Ion Import Across Plasma Membran	−0.22732193	33	6.269e−06	9.293e−04	TRPM1:5 CACNA1B:197 SCN2A:240 CACNA1F:451 SLC24A4:460 SLC24A2:882
Double−Strand Break Repair (GO:0006302)	0.10850866	146	6.383e−06	9.293e−04	ERCC5:32 RNF169:53 ANKLE1:65 RNF168:136 REC8:269 BRCA1:275
Cell Junction Assembly (GO:0034329)	−0.14521986	80	7.328e−06	1.014e−03	SHANK2:11 DRD2:28 SDK1:96 ITGA6:188 DBNL:249 SLITRK1:272
Fatty Acid Catabolic Process (GO:0009062	0.17038062	58	7.338e−06	1.014e−03	PHYH:16 ACADM:17 CPT2:116 HAO2:207 EHHADH:216 ECI2:304
Monoatomic Ion Transport (GO:0006811)	−0.12768593	103	7.855e−06	1.058e−03	SLC15A1:194 SLC12A5:256 CNGB3:274 CHRNA4:285 SLC24A4:460 TRPM3:465

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_NEURONAL_SYSTEM	−0.15397617	352	4.750e−23	3.081e−19	SHANK2:11 GABRR1:14 ADCY1:39 NBEA:48 KCNA10:51 RPS6KA2:63
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	−0.14559047	217	1.652e−13	5.359e−10	GABRR1:14 ADCY1:39 NBEA:48 RPS6KA2:63 PRKCG:91 SLC18A2:95
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.30619674	47	3.873e−13	8.375e−10	NDUFB8:76 ECSIT:170 NDUFV3:281 TMEM126B:302 NDUFB6:335 NDUFA6:471
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_	−0.16609748	159	5.362e−13	8.696e−10	GABRR1:14 ADCY1:39 NBEA:48 RPS6KA2:63 PRKCG:91 CAMK2B:110
REACTOME_COMPLEX_I_BIOGENESIS	0.29877596	48	8.144e−13	1.057e−09	NDUFB8:76 ECSIT:170 NDUFA9:278 NDUFV3:281 TMEM126B:302 NDUFB6:335
REACTOME_MUSCLE_CONTRACTION	−0.15267498	176	3.076e−12	3.326e−09	ATP2A2:67 CAMK2B:110 ITPR1:150 TNNI2:222 SCN2A:240 CAMK2D:287
REACTOME_DEVELOPMENTAL_BIOLOGY	−0.06865569	792	7.478e−11	6.930e−08	DSG3:27 COL3A1:50 TRIO:52 PTK2:54 L1CAM:57 DOCK1:61
KEGG_CALCIUM_SIGNALING_PATHWAY	−0.14673418	161	1.416e−10	1.149e−07	ADCY1:39 ATP2A2:67 PRKCG:91 P2RX2:107 CAMK2B:110 ITPR1:150
REACTOME_INFECTIOUS_DISEASE	−0.07231500	671	2.305e−10	1.523e−07	ADCY1:39 PTK2:54 DOCK1:61 DYNC1H1:80 ACE2:84 ITPR1:150
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	0.20636552	79	2.348e−10	1.523e−07	NDUFB8:76 LRPPRC:106 ECSIT:170 NDUFA9:278 NDUFV3:281 TMEM126B:302
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	0.17825273	99	9.180e−10	5.414e−07	NDUFB8:76 ATP5PO:105 LRPPRC:106 ECSIT:170 NDUFA9:278 NDUFV3:281
REACTOME_GENE_SILENCING_BY_RNA	−0.19234504	82	1.781e−09	9.627e−07	MOV10L1:20 PIWIL1:36 FKBP6:59 TNRC6B:147 ASZ1:237 DICER1:339
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	−0.07730675	514	2.454e−09	1.225e−06	DRD2:28 CADM3:55 NETO1:58 CPZ:60 FAM163B:65 GPR26:79
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	−0.27817786	38	2.985e−09	1.300e−06	TTYH3:138 MAFK:184 INTS1:236 LFNG:332 GET4:413 EIF3B:518
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	−0.10713848	260	3.006e−09	1.300e−06	GABRR1:14 DRD2:28 MAS1:33 HRH3:53 GRM6:85 P2RX2:107
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	−0.08141721	436	6.583e−09	2.669e−06	COL3A1:50 TRIO:52 PTK2:54 L1CAM:57 DOCK1:61 RPS6KA2:63
WP_CALCIUM_REGULATION_IN_CARDIAC_CELLS	−0.15652764	114	8.079e−09	3.083e−06	ADCY1:39 ATP2A2:67 GJA8:75 PRKCG:91 CAMK2B:110 ITPR1:150
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	0.23485170	50	9.317e−09	3.358e−06	NSD3:59 IDO2:60 RAB11FIP1:73 ADAM9:103 DKK4:153 TACC1:161
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	−0.12067316	186	1.472e−08	5.025e−06	SLBP:116 RBM42:161 CPSF7:207 SYMPK:275 NCBP1:292 PAPOLA:293
BENPORATH_ES_WITH_H3K27ME3	−0.05541736	913	1.932e−08	5.967e−06	NKPD1:24 HSF4:42 DGKI:44 VAX2:45 ENTPD2:49 IGFBP1:62
WP_OXIDATIVE_PHOSPHORYLATION	0.23955179	46	1.918e−08	5.967e−06	NDUFB8:76 ATP5PO:105 NDUFA9:278 NDUFV3:281 NDUFB6:335 NDUFA6:471
REACTOME_KERATINIZATION	−0.17802238	83	2.116e−08	6.238e−06	DSG3:27 DSC2:190 KRT20:218 DSC3:259 KLK13:277 PKP3:423
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	−0.10404111	245	2.218e−08	6.256e−06	ADCY1:39 NBEA:48 GRM6:85 PRKCG:91 GNAT2:93 OPN3:108
BENPORATH_EED_TARGETS	−0.05648250	860	2.578e−08	6.967e−06	NKPD1:24 HSF4:42 DGKI:44 VAX2:45 KCNC2:78 SORCS2:86
REACTOME_GABA_RECEPTOR_ACTIVATION	−0.24585341	42	3.561e−08	9.240e−06	GABRR1:14 ADCY1:39 GABRB3:146 KCNJ5:168 KCNJ9:169 ADCY9:348
REACTOME_SIGNALING_BY_GPCR	−0.06629236	596	3.981e−08	9.933e−06	TIAM2:26 DRD2:28 ADCY1:39 DGKI:44 NBEA:48 TRIO:52
WP_2Q37_COPY_NUMBER_VARIATION_SYNDROME	−0.15526161	104	4.619e−08	1.110e−05	KIF1A:72 ASB1:128 KLHL30:181 TRAF3IP1:187 GPC1:317 HDAC4:438
REACTOME_CARDIAC_CONDUCTION	−0.14789741	113	5.793e−08	1.342e−05	ATP2A2:67 CAMK2B:110 ITPR1:150 SCN2A:240 CAMK2D:287 NOS1:341
WONG_MITOCHONDRIA_GENE_MODULE	0.11353669	192	6.163e−08	1.379e−05	CHDH:49 NDUFB8:76 ATP5PO:105 LRPPRC:106 CRYZ:107 SLC16A5:237
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.14275234	120	6.860e−08	1.483e−05	HRH3:53 ZBTB46:106 SAMD10:109 KCNQ2:125 OPRL1:175 TCEA2:247
BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_I	0.12363118	159	7.833e−08	1.612e−05	KIF18B:85 FANCG:160 CLSPN:163 SPAG5:176 ZWINT:187 UHRF1:190
REACTOME_ION_HOMEOSTASIS	−0.23137330	45	7.951e−08	1.612e−05	ATP2A2:67 CAMK2B:110 ITPR1:150 CAMK2D:287 NOS1:341 ATP2B3:492
BENPORATH_SUZ12_TARGETS	−0.05458051	852	8.466e−08	1.664e−05	HSF4:42 DGKI:44 VAX2:45 CADM3:55 KCNC2:78 EGFLAM:81
REACTOME_MITOCHONDRIAL_TRANSLATION	0.15907454	91	1.603e−07	3.001e−05	PTCD3:56 MRPL37:412 MRPS30:469 MRPS23:559 MRPL9:732 MTIF2:831
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	−0.17391663	76	1.619e−07	3.001e−05	DSG3:27 DSC2:190 KRT20:218 DSC3:259 KLK13:277 PKP3:423
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	−0.08673217	308	1.815e−07	3.271e−05	HRH3:53 CADM3:55 FAM163B:65 EGFLAM:81 ADGRB1:89 KCNQ2:125
REACTOME_HCMV_EARLY_EVENTS	−0.16222928	86	2.034e−07	3.566e−05	DYNC1H1:80 NUP37:400 TUBA1B:555 NUP43:620 NUP160:699 NUP133:768
KIM_ALL_DISORDERS_CALB1_CORR_UP	−0.07230861	442	2.143e−07	3.659e−05	KHDRBS3:10 ATP2A2:67 DYNC1H1:80 NPTX1:88 CAMK2B:110 DCLK1:142
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	−0.37959733	15	3.579e−07	5.805e−05	EP400:23 P2RX2:107 SFSWAP:223 POLE:380 GOLGA3:435 GALNT9:551
REACTOME_VOLTAGE_GATED_POTASSIUM_CHANNEL	−0.22983191	41	3.568e−07	5.805e−05	KCNA10:51 KCNC2:78 KCNQ3:100 KCNQ2:125 KCNS3:129 KCNV2:234

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Schizophrenia	−0.05243317	1513	3.933e−11	3.857e−07	TRPM1:5 SHANK2:11 GABRR1:14 DRD2:28 ACTL7A:31 ADCY1:39
Mitochondrial Diseases	0.10374713	335	8.837e−11	4.332e−07	NDUFB8:76 LRPPRC:106 CRYZ:107 CPT2:116 ECSIT:170 CEP89:222
Autism Spectrum Disorders	−0.07875925	456	1.100e−08	3.595e−05	SHANK2:11 DRD2:28 JAKMIP1:46 NBEA:48 TRIO:52 L1CAM:57
Increased CSF lactate	0.20159533	53	3.923e−07	7.825e−04	LRPPRC:106 NDUFA9:278 COX10:286 TMEM126B:302 NDUFS3:597 NDUFS6:625
Nuclear cataract	−0.25761561	32	4.624e−07	7.825e−04	CRYBB2:6 CRYAA:12 CRYBB1:40 HSF4:42 CRYBB3:66 GJA8:75
Primary microcephaly	0.14817066	97	4.788e−07	7.825e−04	ERCC5:32 FANCM:137 FANCG:160 TRMT10A:210 FANCE:239 CENPJ:265
CATARACT, AUTOSOMAL DOMINANT	−0.44923205	10	8.706e−07	1.219e−03	CRYAA2:4 CRYBB2:6 CRYAA:12 CRYBB1:40 GJA8:75 MIP:126
Cataract, Pulverulent	−0.46800229	9	1.164e−06	1.427e−03	CRYBB2:6 CRYBB1:40 HSF4:42 GJA8:75 BFSP2:215 VIM:395
Difficulties with night vision	−0.13298068	109	1.689e−06	1.840e−03	PDE6B:2 TRPM1:5 CDHR1:15 NR2E3:74 GRM6:85 CACNA2D4:112
Abnormality of macular pigmentation	−0.29578100	21	2.718e−06	2.220e−03	PDE6B:2 TRPM1:5 GRM6:85 CACNA2D4:112 PRPH2:193 ABCA4:208
Macular pigmentary changes	−0.29578100	21	2.718e−06	2.220e−03	PDE6B:2 TRPM1:5 GRM6:85 CACNA2D4:112 PRPH2:193 ABCA4:208
Nuclear non−senile cataract	−0.25278563	29	2.480e−06	2.220e−03	CRYBB2:6 CRYAA:12 CRYBB1:40 CRYBB3:66 GJA8:75 MIP:126
Comatose	0.15935214	71	3.528e−06	2.471e−03	ACADM:17 CPT2:116 TMEM126B:302 HMGCL:417 ABCD2:517 NDUFS3:597
Night Blindness	−0.12608336	114	3.466e−06	2.471e−03	PDE6B:2 TRPM1:5 CDHR1:15 NR2E3:74 GRM6:85 CACNA2D4:112
Congenital cataract	−0.14135843	88	4.726e−06	3.089e−03	CRYAA2:4 CRYBB2:6 CRYAA:12 CRYBB1:40 HSF4:42 CRYBB3:66
CATARACT, COPPOCK−LIKE	−0.43910381	9	5.081e−06	3.114e−03	CRYBB2:6 CRYBB1:40 GJA8:75 BFSP2:215 VIM:395 GJA3:1031
Channelopathies	−0.19573689	45	5.630e−06	3.247e−03	KCNJ5:168 CNGB3:274 KCNJ6:374 CACNA1F:451 RYR2:505 SHANK3:572
Mouth Neoplasms	−0.12981944	99	8.353e−06	4.550e−03	PTK2:54 PTH1R:178 VIM:395 TIAM1:459 MGMT:511 HMGCS2:515
Severe myopia	−0.13528396	88	1.187e−05	6.126e−03	PDE6B:2 TRPM1:5 GRM6:85 CACNA2D4:112 CABP4:235 CNGB3:274
Night blindness, congenital stationary	−0.26218851	23	1.352e−05	6.446e−03	PDE6B:2 TRPM1:5 USP11:30 GRM6:85 CACNA2D4:112 ABCA4:208
Alzheimer's Disease	−0.03413572	1551	1.381e−05	6.446e−03	SHANK2:11 NKPD1:24 DRD2:28 PLD1:34 COL3A1:50 HRH3:53
NADH:Q(1) Oxidoreductase deficiency	0.24979551	25	1.547e−05	6.894e−03	TMEM126B:302 NDUFS3:597 NDUFS6:625 FOXRED1:638 NDUFAF2:701 TIMMDC1:858
Congenital myopathy (disorder)	−0.12369091	99	2.180e−05	8.907e−03	COL12A1:92 BIN1:124 TNNI2:222 COL6A2:372 HACD1:404 DES:443
Lipoprotein (a) measurement	−0.34080717	13	2.098e−05	8.907e−03	SYTL3:254 SLC22A3:566 PRKN:943 DSCAML1:1136 SERAC1:1221 TAGAP:1324
Epilepsy, Temporal Lobe	−0.09840491	154	2.634e−05	9.934e−03	PLD1:34 KCNQ3:100 BIN1:124 GABRB3:146 GRM2:220 SCN2A:240
Skin Diseases, Genetic	−0.20844649	34	2.619e−05	9.934e−03	ATP2A2:67 ITGA6:188 ITGB4:261 IGFBP3:494 COL7A1:588 KRT5:644
Alcoholic Intoxication, Chronic	−0.06551514	351	2.797e−05	1.016e−02	GABRR1:14 DRD2:28 ADCY1:39 SLC18A2:95 CNTN4:105 GABRB3:146
Color vision defect, severe	−0.19916614	36	3.577e−05	1.169e−02	CDHR1:15 GNAT2:93 CACNA2D4:112 PRPH2:193 ABCA4:208 CNGB3:274
Abnormal color vision	−0.19916614	36	3.577e−05	1.169e−02	CDHR1:15 GNAT2:93 CACNA2D4:112 PRPH2:193 ABCA4:208 CNGB3:274
Loss in color vision	−0.19916614	36	3.577e−05	1.169e−02	CDHR1:15 GNAT2:93 CACNA2D4:112 PRPH2:193 ABCA4:208 CNGB3:274
Cataract, Central Saccular, With Sutural	−0.48611226	6	3.730e−05	1.180e−02	CRYBB2:6 GJA8:75 MIP:126 BFSP2:215 CRYBA1:219 CRYGS:538
Seizures, Focal	−0.10825723	120	4.362e−05	1.203e−02	DRD2:28 DYNC1H1:80 KCNQ3:100 KCNQ2:125 SLC6A2:203 SCN2A:240
Charcot−Marie−Tooth Disease	−0.07656727	244	4.117e−05	1.203e−02	L1CAM:57 SBF1:70 KIF1A:72 DYNC1H1:80 PRKCG:91 ITPR1:150
MITOCHONDRIAL COMPLEX I DEFICIENCY	0.23150463	26	4.415e−05	1.203e−02	TMEM126B:302 NDUFS3:597 NDUFS6:625 FOXRED1:638 NDUFAF2:701 TIMMDC1:858
Renal salt wasting	0.30580842	15	4.129e−05	1.203e−02	CLCNKA:41 CLCNKB:86 SARS2:248 KCNJ1:555 BSND:623 CYP11A1:1236
Seckel syndrome	0.21254266	31	4.241e−05	1.203e−02	DNMT1:89 CENPJ:265 BRCA1:275 DNA2:577 PLK4:581 NIN:720
Acute necrotizing encephalopathy	0.26718966	19	5.551e−05	1.471e−02	TMEM126B:302 NDUFS3:597 NDUFS6:625 FOXRED1:638 NDUFAF2:701 TIMMDC1:858
Epidermolysis Bullosa Simplex	−0.26657309	19	5.775e−05	1.490e−02	ITGB4:261 PLEC:373 VIM:395 DES:443 KRT5:644 FN1:654
Cataract microcornea syndrome	−0.38593724	9	6.095e−05	1.494e−02	CRYBB2:6 CRYAA:12 CRYBB1:40 GJA8:75 CRYBA4:396 CRYGD:1184
Hyponatremia	0.19083312	37	5.951e−05	1.494e−02	CLCNKA:41 CLCNKB:86 NFKB2:204 SARS2:248 SAMD9:272 CLDN16:293

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0003635 abnormal synaptic transmissio	−0.11627705	390	2.182e−14	1.037e−11	DRD2:28 MAS1:33 ADCY1:39 NBEA:48 HRH3:53 L1CAM:57
MP0005551 abnormal eye electrophysiolog	−0.14809037	136	3.622e−09	8.602e−07	PDE6B:2 CPLX3:3 TRPM1:5 GABRR1:14 CDHR1:15 NR2E3:74
MP0005253 abnormal eye physiology	−0.15323306	89	6.951e−07	1.101e−04	TRPM1:5 NR2E3:74 GJA8:75 EGFLAM:81 MIP:126 PRPH2:193
MP0009745 abnormal behavioral response	−0.10188552	190	1.826e−06	2.168e−04	DRD2:28 KCNC2:78 SLC18A2:95 KCNQ2:125 GABRB3:146 HOMER2:174
MP0002398 abnormal bone marrow	0.05207004	750	5.203e−06	3.536e−04	JAK3:5 AFF1:8 THPO:26 NEDD4:43 DST:54 FOXN1:104
MP0002063 abnormal learning/memory/cond	−0.07501150	329	5.211e−06	3.536e−04	DRD2:28 ADCY1:39 HRH3:53 L1CAM:57 NETO1:58 SLC18A2:95
MP0002272 abnormal nervous system	−0.10192790	175	4.463e−06	3.536e−04	GABRR1:14 DRD2:28 KCNC2:78 KCNQ3:100 KCNQ2:125 GABRB3:146
MP0002064 seizures	−0.08744655	234	6.057e−06	3.596e−04	DRD2:28 NETO1:58 KCNC2:78 KCNQ3:100 KCNQ2:125 GABRB3:146
MP0002572 abnormal emotion/affect behav	−0.07146219	276	6.391e−05	3.373e−03	DRD2:28 MAS1:33 ADCY1:39 HRH3:53 L1CAM:57 KCNC2:78
MP0010094 abnormal chromosome stability	0.15022635	57	9.375e−05	4.453e−03	FOXN1:104 FANCG:160 BRCA1:275 BARD1:564 BUB1B:637 MBD1:756
MP0002066 abnormal motor capabilities/c	−0.03897226	994	1.375e−04	5.025e−03	RPTOR:13 DRD2:28 VAX2:45 NBEA:48 HRH3:53 L1CAM:57
MP0002135 abnormal kidney morphology	0.05660375	420	1.197e−04	5.025e−03	NAGLU:71 SLC2A10:100 RBPJ:211 GAS6:253 CUBN:254 GCNT2:257
MP0001968 abnormal touch/ nociception	−0.19916550	31	1.286e−04	5.025e−03	OPRL1:175 SLC12A5:256 GLRA3:257 P2RX3:335 TACR1:1379 HMOX2:1489
MP0002429 abnormal blood cell	0.03668152	1131	1.683e−04	5.712e−03	JAK3:5 AFF1:8 THPO:26 NEDD4:43 NAGLU:71 FOXN1:104
MP0000689 abnormal spleen morphology	0.04989877	510	2.155e−04	6.823e−03	JAK3:5 ERCC5:32 DNMT1:89 SLC2A10:100 FOXN1:104 MPL:132
MP0006072 abnormal retinal apoptosis	−0.18773120	32	2.457e−04	7.295e−03	NR2E3:74 PRPH2:193 CNGB3:274 POU4F2:704 LAMB2:896 FOS:1020
MP0000598 abnormal liver morphology	0.05232840	443	2.693e−04	7.525e−03	PHYH:16 ERCC5:32 NAGLU:71 ANGPTL3:92 FOXN1:104 GTF2IRD1:200
MP0001970 abnormal pain threshold	−0.08773238	144	3.256e−04	8.591e−03	GABRR1:14 ADCY1:39 KIF1A:72 OPRL1:175 TSHZ3:179 CACNA1B:197
MP0002085 abnormal embryonic tissue	0.04415929	585	5.054e−04	1.263e−02	TRPM6:1 NSD1:81 CHRD:96 RRN3:112 ARL13B:127 SHROOM3:152
MP0002067 abnormal sensory capabilities	−0.06573374	238	6.069e−04	1.441e−02	ATP2A2:67 GABRB3:146 ITPR1:150 OPRL1:175 CACNA1B:197 DBNL:249
MP0005332 abnormal amino acid	0.09255162	113	7.536e−04	1.705e−02	ACADM:17 GCNT2:257 ACADVL:459 NEK8:546 BCAT2:895 CTH:898
MP0002876 abnormal thyroid physiology	0.23371095	17	8.620e−04	1.861e−02	RUNX2:735 HPN:1109 GFRA4:1591 GLIS3:1596 CGA:2490 THRB:2564
MP0005193 abnormal anterior eye	−0.06400166	225	1.154e−03	2.236e−02	CRYBB2:6 CRYAA:12 DSG3:27 HSF4:42 GJA8:75 MIP:126
MP0009642 abnormal blood homeostasis	0.03091357	1196	1.224e−03	2.236e−02	TRPM6:1 PHYH:16 ACADM:17 MBOAT4:34 ANGPTL3:92 CHRD:96
MP0004859 abnormal synaptic plasticity	−0.17408832	29	1.204e−03	2.236e−02	ADCY1:39 CAMK2B:110 CAMK2A:920 GRIN1:1628 FOXP2:1697 GRIN2B:1965
MP0002084 abnormal developmental patter	0.05469352	314	1.152e−03	2.236e−02	NSD1:81 DNMT1:89 CHRD:96 RRN3:112 ARL13B:127 ECSIT:170
MP0000609 abnormal liver physiology	0.05065209	360	1.338e−03	2.271e−02	PHYH:16 ACADM:17 ANGPTL3:92 SOAT2:164 CPT1C:179 GTF2IRD1:200
MP0001486 abnormal startle reflex	−0.09876593	90	1.299e−03	2.271e−02	DRD2:28 CACNA1B:197 LOXHD1:231 NTNG2:354 MID1:581 ABCA2:648
MP0005380 embryogenesis phenotype	0.04923080	373	1.530e−03	2.422e−02	NLRP5:63 RLIM:93 RRN3:112 SHROOM3:152 AFF4:169 UHRF1:190
MP0001672 abnormal embryogenesis/ devel	0.04923080	373	1.530e−03	2.422e−02	NLRP5:63 RLIM:93 RRN3:112 SHROOM3:152 AFF4:169 UHRF1:190
MP0005369 muscle phenotype	−0.06649474	192	1.748e−03	2.615e−02	RPTOR:13 SH3BP2:38 DOCK1:61 CAPN3:73 SYNE1:144 COL15A1:299
MP0002735 abnormal chemical nociception	−0.17112174	28	1.762e−03	2.615e−02	ADCY1:39 OPRL1:175 CACNA1B:197 P2RX3:335 PIRT:507 ADCY8:782
MP0008058 abnormal DNA repair	0.14706360	37	2.021e−03	2.889e−02	TEX15:245 POLK:360 ERCC1:1205 APC:1702 POLL:1829 BRCA2:1911
MP0005391 vision/eye phenotype	−0.06732201	181	2.068e−03	2.889e−02	TRPM1:5 NR2E3:74 GJA8:75 EGFLAM:81 MIP:126 PRPH2:193
MP0008875 abnormal xenobiotic pharmacok	0.22072454	16	2.264e−03	3.010e−02	CYP1A1:333 CYP1A2:365 MTF1:695 AS3MT:1403 AHR:1900 COMT:2271
MP0002233 abnormal nose morphology	0.10115510	77	2.281e−03	3.010e−02	CHRD:96 ALDH1A1:114 MKKS:273 B3GNT2:565 HESX1:653 PAX3:703
MP0000003 abnormal adipose tissue	0.15550015	32	2.388e−03	3.066e−02	ACADVL:459 PPARD:655 ENPP1:970 PGF:1513 ARID5B:1822 LPL:1825
MP0000716 abnormal immune system	0.03182048	922	2.489e−03	3.111e−02	JAK3:5 AFF1:8 NEDD4:43 NAGLU:71 FOXN1:104 MPL:132
MP0002736 abnormal nociception after	−0.18725535	21	3.016e−03	3.674e−02	GLRA3:257 TACR1:1379 HMOX2:1489 SCN10A:2084 ESR1:2260 PROKR1:2289
MP0002752 abnormal somatic nervous	−0.03912237	533	3.094e−03	3.674e−02	PDE6B:2 CPLX3:3 VAX2:45 L1CAM:57 NR2E3:74 EGFLAM:81

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Brodmann.area.9	−0.14990244	162	6.574e−11	3.550e−09	MAS1:33 TMEM132D:41 NETO1:58 KCNC2:78 GPR26:79 KCNQ3:100
liver	0.10540606	322	1.600e−10	4.320e−09	ACY1:21 THPO:26 SERPINA7:35 ACOX2:38 IDO2:60 ANGPTL3:92
EBV.lymphocyte	0.06632896	652	2.407e−08	4.333e−07	SMC4:9 ACADM:17 DLGAP5:57 ANKLE1:65 KIF18B:85 TMEM131L:87
cerebral.cortex	−0.15005435	103	1.648e−07	2.225e−06	MAS1:33 FAM163B:65 KIF1A:72 ADGRB1:89 OPRL1:175 MCHR1:497
cerebellar.hemisphere	−0.05975982	459	1.841e−05	1.988e−04	PHTF1:18 ADCY1:39 VAX2:45 NBEA:48 CADM3:55 NPTX1:88
pancreas	0.12373404	93	4.043e−05	3.638e−04	CPB1:2 TMED6:90 ERP27:113 CELA3B:151 KCNK16:263 CTRC:289
spleen	0.06753213	291	9.440e−05	7.282e−04	AKNA:7 DENND1C:80 EXOC3L1:88 SOAT2:164 CLEC4F:183 LRRC25:262
skeletal.muscle	−0.07055033	232	2.537e−04	1.712e−03	ATP2A2:67 CAPN3:73 BIN1:124 CAVIN4:145 KLHL30:181 ABLIM2:210
cortex.kidney	0.09803235	110	4.114e−04	2.468e−03	SLC5A12:27 CLCNKB:86 CUBN:254 PKHD1:287 CLDN16:293 DCDC2:383
vagina	−0.10163977	89	9.679e−04	5.227e−03	DSG3:27 PGLYRP4:76 SLC18A2:95 DSC2:190 DSC3:259 TMPRSS11F:357
cerebellum	−0.04981913	354	1.569e−03	7.702e−03	PHTF1:18 HRH3:53 L1CAM:57 ZBTB46:106 KCNQ2:125 MAPK8IP2:132
blood	0.05619617	266	1.856e−03	8.352e−03	JAK3:5 BPI:138 RHAG:215 TREML2:228 EPB42:231 DHX34:238
nucleus.accumbens	−0.09212331	93	2.241e−03	9.308e−03	JAKMIP1:46 INSYN2A:71 PRKCG:91 PNPLA5:464 CPNE5:466 SHISA7:500
suprapubic.skin	−0.06444814	181	3.043e−03	1.174e−02	NKPD1:24 DSG3:27 PGLYRP4:76 DSC3:259 PARD6G:294 TMPRSS11F:357
leg.skin	−0.05483334	209	6.860e−03	2.470e−02	TRPM1:5 CDHR1:15 NKPD1:24 DSG3:27 PGLYRP4:76 TRIM29:77
hypothalamus	−0.09427441	58	1.327e−02	4.479e−02	GLRA3:257 DNER:340 MC4R:361 GRM7:405 PIRT:507 GABRG3:662
fallopian.tube	0.09889466	48	1.803e−02	5.728e−02	ADAMTS3:19 CCDC17:42 ADGRG4:172 GMNC:208 GREB1L:953 GASK1A:1253
thyroid	0.05512645	147	2.201e−02	6.604e−02	TLE6:6 CLCNKA:41 PRTG:45 IDO2:60 WDR72:168 PRSS16:184
Brodmann.area.24	−0.10976944	29	4.103e−02	1.166e−01	DRD5:363 FAM189A1:371 SLC17A7:979 CRH:1077 KCNS1:1097 TAC3:1491
putamen	−0.15318984	14	4.732e−02	1.278e−01	DIPK1C:954 PDE10A:1256 LRRC10B:1966 GPR88:2196 RGS9:3006 KCNH4:3647
hippocampus.proper	−0.18881403	9	4.990e−02	1.283e−01	SLC17A7:979 NEUROD6:1683 CNIH2:2374 OLIG2:3787 FEZF2:3994 GRP:5370
esophagus.mucosa	−0.03714973	213	6.452e−02	1.584e−01	DSG3:27 PGLYRP4:76 DSC2:190 FZD10:225 DSC3:259 KLK13:277
minor.salivary.gland	0.06200320	69	7.590e−02	1.782e−01	TCN1:361 BPIFB1:431 ASCL3:543 PAX3:703 EHF:817 MMP27:1018
subcutaneous.adipose	0.05777437	71	9.345e−02	2.103e−01	SIGLEC1:155 GPBAR1:192 LVRN:264 CYP1A1:333 ITIH5:846 MEOX1:872
coronary.artery	0.12149034	14	1.157e−01	2.500e−01	SUSD5:24 SOST:1624 MYO18B:1787 HEYL:2435 CCN3:3031 GJA5:3289
C1.spinal.cord	−0.03443678	171	1.235e−01	2.565e−01	PDE6B:2 TTYH2:37 C10orf90:82 SEC14L5:101 MBP:315 GPR62:533
adrenal.gland	0.04165752	107	1.386e−01	2.582e−01	CYP17A1:126 SLC16A9:166 CELA1:195 SLC14A2:243 DNAH11:432 CYSLTR2:513
sigmoid.colon	−0.06107853	50	1.361e−01	2.582e−01	HDAC4:438 DES:443 GREM2:647 CHRM2:677 TACR2:758 CNN1:849
stomach	0.04793460	84	1.304e−01	2.582e−01	PGC:48 TMED6:90 TMPRSS2:219 PIK3C2G:225 CLDN23:536 ADAM28:583
prostate	−0.06697933	37	1.593e−01	2.805e−01	NR2E3:74 P2RX2:107 KRT5:644 ACP3:830 STEAP2:1166 PSCA:1168
transformed.skin.fibroblast	−0.02643069	243	1.610e−01	2.805e−01	COL3A1:50 COL12A1:92 CNTN3:99 OPN3:108 MMP2:269 RCN3:306
uterus	0.04801809	53	2.277e−01	3.843e−01	CLEC4F:183 TEX15:245 ADAMTS19:492 CHRDL2:606 HOXD4:822 ENPP1:970
atrium.auricle	0.03562271	89	2.475e−01	4.050e−01	TNNI3K:33 XIRP1:558 LMOD3:590 BMP10:656 SMCO1:736 NMRK2:767
breast	−0.08347815	14	2.798e−01	4.317e−01	KRT5:644 PLIN1:1431 TNN:2493 NPY2R:2663 THRSP:4284 LEP:4587
tibial.artery	−0.03159909	101	2.749e−01	4.317e−01	SDK1:96 WFDC1:97 TMEM181:266 VIM:395 PRELP:727 KCNE4:748
amygdala	−0.10623810	8	2.982e−01	4.473e−01	SLC17A7:979 NEUROD6:1683 PTPRZ1:4496 SLC38A8:5351 PCDH15:6687 NEUROD2:867
endocervix	−0.03301949	70	3.411e−01	4.978e−01	SLC18A2:95 TMEM132C:470 ISLR:708 SPON2:775 ELFN1:845 SNED1:1187
caudate.nucleus	−0.04802187	26	3.972e−01	5.645e−01	GABRG1:971 MLC1:1066 LRRC10B:1966 GPR88:2196 AQP4:2207 RLBP1:2241
esophagus.muscularis.mucosa	0.04532261	26	4.243e−01	5.875e−01	PRUNE2:353 F2RL2:487 CCDC69:833 GADL1:1021 SMTN:2109 RGMB:2947
bladder	−0.04551371	23	4.504e−01	5.932e−01	PADI3:410 KRT13:1121 PSCA:1168 PLA2G2F:1428 RD3:3270 TMC7:3300